QIYUAN XU

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EDUCATION

Zhejiang U. Chu Kochen Honors College B.E. in Computer Science *Sep.2013 - Jun.2017*
Zhejiang U. Master in Cyber Security *Sep.2017 - Jun.2020*
Nanyang Technological U. PhD student, supervised by Conrad Watt *Aug.2022 - Now*

Awards
Second-Class Scholarship for Elite Students in Basic Science, Zhejiang U.
Scholarship for Excellence in Research and Innovation, Zhejiang U.
Academic Scholarship, Zhejiang U.

PUBLICATIONS

- AoA: Theorem Proving Agent over Abstract Syntax Tree of Redesigned Language.
  Qiyuan Xu, Joshua Ong, Renxi Wang, Wenda Li, Haonan Li, Luke Ong, Conrad Watt. Accepted in **ICSE'27**, CORE ranking **A***, CCF ranking **A**.
- Neural Theorem Proving for Verification Conditions: A Real-World Benchmark.
  Qiyuan Xu, Xiaokun Luan, Renxi Wang, Joshua Ong, Peixin Wang, Haonan Li, Wenda Li, Conrad Watt. Published in **ICLR'26**, CORE ranking **A***, CCF ranking **A**.
- A Minimalist Proof Language for Neural Theorem Proving over Isabelle/HOL.
  Qiyuan Xu, Renxi Wang, Peixin Wang, Haonan Li, Conrad Watt.
  Published in **OOPSLA'26**, CORE ranking **A**, CCF ranking **A**
- Generically Automating Separation Logic by Functors, Homomorphisms and Modules.
  Qiyuan Xu, David Sanan, Zhe Hou, Xiaokun Luan, Conrad Watt, Yang Liu.
  Published in **POPL'25**, CORE ranking **A***, CCF ranking **A**
- Why the Proof Fails in Different Versions of Theorem Provers: An Empirical Study of Compatibility Issues in Isabelle. Xiaokun Luan, David Sanan, Hou Zhe, **Qiyuan Xu**, Chengwei Liu, Yufan Cai, Yang Liu and Meng Sun. Published in **FSE'25**, CORE ranking **A***, CCF ranking **A**

REMARKABLE EVENT

As a key member in the team, established a new **world record** on 3KW Linpack testing, during the ASC Student Supercomputer Challenge 2016. See official report.

COMPETITION HONORS

ASC15 (ASC Student Supercomputer Challenge 2015) First Prize
ASC16 First Class Award ASC16 Highest LINPACK

PATENTS

- coinventor, CN109753288A, *Intelligent contract compiling method suitable for formal verification*
- coinventor, CN109918375A, *Large text storage, indexing and retrieval method based on block chain and distributed storage*
- coinventor, CN111311255A, *Intelligent contract formal verification and error correction method based on oracle machine*

ENGINEERING & RESEARCH EXPERIENCE

Research Associate & PhD student in **Nanyang Technology University** *2022 - now.*
Supervisor: Conrad Watt, assistant professor, NTU, conrad.watt@ntu.edu.sg

Project: *Phi-System*, a certified programming language & program verification framework built on top of Isabelle/HOL. It is based on a first-order fictional separation logic, equipped with an algebra-based generic automation mechanism and a symbolic execution engine for assisting certified programming.
Projects: Isabelle infrastructures for Neural Theorem Proving.

Senior engineer in **Hangzhou Yunphant Network Technology Co.,Ltd** *2020 - 2022.*
Director: Dr. Butian Huang, founder & CEO, Yunphant Co.,Ltd, hr@yunphant.com

Project: Blockchain-based Manufacturing Service Supply and Demand Matching Platform ‖ Group member.
Key R&D Program of Zhejiang Province, No. 2021C01116
Advisor: Dr. Zhenguang Liu, Zhejiang U.
Audited key smart contracts, and found several deficiencies including a critical re-entry vulnerability resulting in stealing of total assets potentially.

Graduate student in **the lab of Very Large Information System, Zhejiang U.** *2017-2020*
Advisor: Dr. Liang Cai, Vice Dean of the Software College, Zhejiang U., leoncai@zju.edu.cn.
Project: GPU based implementation of SM2 cyptographic algorithm.

- Implemented finite group operations of elliptical curves on the CUDA device, with hotpot written in manually optimized PTX assemble code.
- The work has been applied in the industrial product of Hyperchain, and its productive distribution for the Shanghai Stock Exchange.

Key member in **Supercomupting Team of Zhejiang University** *2014 - 2017.*
Advisor: Associate Prof. Jianhai Chen, Zhejiang U., chenjh919@zju.edu.cn.
Project 1: Parallel optimization – MASNUM (marine science and numerical modeling)

- Implemented the first GPU-accelerated version of the MASNUM software, during the ASC16. The optimized software reached the theoretical performance peak of the GPU device, Nvidia Tesla K40.

Project 2: SKA (Square Kilometer Array telescope) data processing application Gridding

- Implemented a CPU-based cluster-scale parallel version of the hotpot of the SKA-Gridding, which is a 2-dim convolution, based on the MPI and OpenMP, specified for Intel Many Integrated Core.

OTHER SOFTWARES

- Isabelle REPL, Isabelle-MCP, basic infrastructures for AI-based theorem proving, allowing machine learning systems to connect the Isabelle theorem prover.
- Verification for Uniswap v3, the core functions of the smart contract Uniswap v3.